

Ex no:1

Date : 30/7/26

## STUDY OF VARIOUS COMMANDS USED IN LINUX AND WINDOWS

### AIM:

To study about various networking commands on Linux operating system.

### BASIC LINUX COMMANDS

#### 1. pwd

**Description:** Points the current working directory.

Output:

```
$ pwd  
  
/home/liveuser
```

#### 2. mkdir Bedora

**Description:** Create a new directory.

Output:

```
$ mkdir Bedora  
  
$ ls  
  
Desktop  Documents  Downloads  Bedora
```

#### 3. cd Bedora

**Description:** Changes the current directory.

Output:

```
$ cd Bedora  
  
$ pwd  
  
/home/liveuser/Bedora
```

#### 4. touch example.txt

**Description:** Create an empty file.

Output:

```
$ touch example.txt  
  
$ ls  
  
example.txt
```

### 5. **echo "Hi" > example.txt**

**Description:** Write text into a file.

Output:

```
$ echo "Hi" > example.txt
```

### 6. **cat example.txt**

**Description:** Display the contents of a file.

Output:

```
$ cat example.txt
```

```
Hi
```

### 7. **grep "Hi" example.txt**

**Description:** Searches for a specific pattern in a file.

Output:

```
$ grep "Hi" example.txt
```

```
Hi
```

### 8. **wc example.txt**

**Description:** Display the number of lines, words and bytes in a file.

Output:

```
$ wc example.txt
```

```
1 1 3 example.txt
```

### 9. **wc -l example.txt**

**Description:** Counts the number of lines in a file.

Output:

```
$ wc -l example.txt
```

```
1 example.txt
```

### 10. **wc -w example.txt**

**Description:** Counts the number of words in a file.

Output:

```
$ wc -w example.txt
```

```
1 example.txt
```

### 11. **wc -c example.txt**

**Description:** Counts the number of bytes in a file.

Output:

```
$ wc -c example.txt
3  example.txt
```

### 12. **wc -m example.txt**

**Description:** Counts the number of characters in a file.

Output:

```
$ wc -m example.txt
3  example.txt
```

### 13. **wc -L example.txt**

**Description:** Displays the length of the longest line in a file.

Output:

```
$ wc -L example.txt
2  example.txt
```

## **BASIC NETWORKING COMMANDS**

### 1. **hostname**

**Description:** Displays the system's hostname.

Output:

```
$ hostname
localhost-live
```

### 2. **route**

**Description:** Displays the routing table.

Output:

```
$ route
Kernel IP routing table
```

### 3. **netstat**

**Description:** Shows network connections and routing information.

Output:

```
$ netstat
```

```
Active Internet connections
```

#### 4. ip address show

**Description:** Displays IP address and network interface details.

Output:

```
$ ip address show
```

```
1: lo:
```

```
2: enp0s3:
```

#### 5. nslookup

**Description:** Finds the IP address of a domain.

Output:

```
$ nslookup www.google.com
```

```
Name: www.google.com
```

```
Address: 142.250.xxx.xxx
```

#### 6. ping

**Description:** Tests connectivity to a host.

Output:

```
$ ping www.google.com
```

```
64 bytes from ...
```

#### 7. arp

**Description:** Displays the ARP cache.

Output:

```
$ arp -a
```

```
gateway (10.0.2.2)
```

#### 8. traceroute

**Description:** Displays the route taken by packets to reach a destination.

Output:

```
$ traceroute www.google.com
```

```
1 gateway
```

2 . . . .

**RESULT:**

The basic Linux and networking commands were executed successfully and expected outputs were obtained.